Gdańsk University of Technology
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Project Process Documentation

Hardware Design and Manufacturing

v1.0

Fundus Camera Prototype

Project Subject:	Glaucoma detection using intelligent analysis of fundus images
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1 Goal and Scope

The primary objective of this development process was to design and manufacture a functional, low-cost fundus camera prototype. The device is intended to serve as a hardware for the core component of the project an application for automated glaucoma detection. Key design constraints included ensuring portability and achieving a significant reduction in production costs compared to professional fundus camera, while maintaining sufficient image quality for diagnostic analysis.

2 Methodology

The project began with a source study, involving the analysis of various existing DIY fundus camera designs. Each approach was evaluated based on the availability of materials, the replicability of the assembly process, and the quality of the resulting retinal images. Following the research phase, an iterative prototyping strategy was adopted. This "trial and error" approach allowed for the continuous refinement of the design, where each stage of assembly was validated against the main goal and the reference materials identified during the research phase.

3 Design Rationale

3.1 Selection of Optical Components

3.2 Selection of Optical Components

Smartphone: An iPhone 14 Pro Max was selected due to its advanced computational photography capabilities and high availability. Furthermore, the choice was driven by the project's software requirements, as the integrated glaucoma detection application was developed natively for the iOS ecosystem, ensuring seamless hardware-software compatibility.

Lens Selection: 20D lens was used, according to chosen camera design video.

3.3 Structural Material Selection

Assembly Interface: A dedicated smartphone case was used as the mounting interface to ensure a repeatableconnection between the camera and the optical tube, allowing for easy removal of the phone.

4 Prototype History and Iterations

4.1 Initial Optical Testing

The preliminary testing phase focused on evaluating the optical behavior of the lens and assessing the practical challenges of retinal imaging. Following the preparation of the tube's interior, the lens was temporarily mounted for initial trials. These tests revealed that structural instability between the components was a significant obstacle. Furthermore, the results highlighted the necessity for precise calibration regarding the positioning of the lens relative to the subject's eye to achieve optimal focus and field of view.

4.2 Structural Optimization and Mounting

During this phase, the tube length was reduced from 25 cm to 20 cm to re-evaluate the focal path, as the initial configuration failed to produce the expected imaging results. Additionally, the tube was joined to the smartphone case, which successfully addressed the previously identified stability issues. Despite this improved structural integrity, internal light reflections blocked the view of the fundus. Consequently, the team concluded that further trials required external expert consultation to overcome these specific optical barriers and achieve the desired project outcomes.

4.3 Consultation with ophthalmologist

The team conducted a technical consultation with an experienced ophthalmologist to address critical design challenges and optical uncertainties. During this session, the physician performed a series of tests using various ophthalmic lenses within a simulated prototype environment. These practical trials, followed by a detailed technical discussion, resulted in several key strategic decisions. It was determined that the current lens configuration required further refinement to optimize the focal plane. Consequently, the ophthalmologist recommended a specialized consultation with an optical physicist to address these adjustments.

4.4 Consultation with an Optical Physicist

The consultation arranged through the ophthalmologist provided the team with critical insights into the optical design. The physicist confirmed the feasibility of developing a functional system utilizing only a single lens. Furthermore, the discussion highlighted the necessity of introducing an adjustable mechanism to control the distance between the lens and the subject's eye, ensuring proper focusing and alignment. Based on these optical requirements, the team decided that the next iteration of the prototype should be manufactured using 3D printing technology to allow for precise, custom geometric adjustments.

4.5 Consultation with a 3D Printing Expert

Following the decision to redesign the hardware, the architecture of the second prototype was reviewed with a recommended manufacturing specialist. The camera assembly was modularized into critical functional components. As a result of this consultation, the team was tasked with developing the 3D models of these components using the Autodesk Fusion 360 environment.

4.6 Final Integration

Ultimately, the team decided to completely abandon the initial DIY concept involving a manual tube setup and a smartphone case. Due to persisting alignment and reflection issues, the original approach proved unviable for a precise medical application. Consequently, the final integration stage focused entirely on pivoting towards a professional, 3D-printed modular design based on the specialized expert feedback.

5 Challenges and Lessons Learned

The primary lesson learned from this iteration was the realization that constructing a stable and reliable optical alignment using simple, homemade materials is practically impossible under domestic conditions. Precise focusing, stabilization, and control over light reflections require a level of geometric accuracy that standard hardware components cannot provide. This critical challenge shifted the team's entire perspective, leading to the strategic decision to adopt 3D printing technology as the only feasible way to manufacture the required components and complete the prototype successfully.